Assignment No : 6

Case Study : Employee Appraisal System

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- **A. Strored Procedures**

1. Total Appraisal Score of Employee

```
mysql> DELIMITER $$
mysql>
mysql> CREATE PROCEDURE GetEmployeeRating(IN empId INT)
    -> BEGIN
    ->     SELECT Emp_ID, AVG(Rating) AS Avg_Rating
    ->     FROM appraisal
    ->     WHERE Emp_ID = empId
    ->     GROUP BY Emp_ID;
    -> END $$
Query OK, 0 rows affected (0.01 sec)
```

Testing:

```
mysql> CALL GetEmployeeRating(101);
+--------+------------+
| Emp_ID | Avg_Rating |
+--------+------------+
|    101 |     4.0000 |
+--------+------------+
1 row in set (0.01 sec)
```

2. Insert Appraisal

```
mysql> DELIMITER $$
mysql>
mysql> CREATE PROCEDURE AddAppraisal(
    ->     IN empId INT,
    ->     IN rating INT,
    ->     IN review VARCHAR(100)
    -> )
    -> BEGIN
    ->     INSERT INTO appraisal (Emp_ID, Rating, Review, Appraisal_Date)
    ->     VALUES (empId, rating, review, CURDATE());
    -> END $$
Query OK, 0 rows affected (0.01 sec)

mysql>
mysql> DELIMITER ;
```

Testing:

```
mysql> CALL AddAppraisal(101, 5, 'Excellent performance this quarter');
Query OK, 1 row affected (0.01 sec)
```

```
mysql> SELECT * FROM appraisal;
+--------------+--------+--------+-----------------------------------+----------------+
| Appraisal_ID | Emp_ID | Rating | Review                            | Appraisal_Date |
+--------------+--------+--------+-----------------------------------+----------------+
|            1 |    101 |      5 | Excellent performance this quarter | 2026-04-14     |
+--------------+--------+--------+-----------------------------------+----------------+
1 row in set (0.00 sec)
```

3. Get High Performance Employees

```
mysql> CREATE PROCEDURE GetTopEmployees()
    -> BEGIN
    ->     SELECT e.Emp_ID, e.Name, AVG(a.Rating) AS Avg_Rating
    ->     FROM employee e
    ->     JOIN appraisal a ON e.Emp_ID = a.Emp_ID
    ->     GROUP BY e.Emp_ID, e.Name
    ->     HAVING AVG(a.Rating) >= 4;
    -> END$$
Query OK, 0 rows affected (0.01 sec)
```

Testing:

```
mysql> CALL GetTopEmployees();
+--------+-------------+------------+
| Emp_ID | Name        | Avg_Rating |
+--------+-------------+------------+
|    101 | Alice Smith |     5.0000 |
+--------+-------------+------------+
1 row in set (0.01 sec)
```

➢ B.Functions

1. Average Rating of All Employees:

```
mysql> DELIMITER $$
mysql>
mysql> CREATE FUNCTION AvgRating()
    -> RETURNS FLOAT
    -> DETERMINISTIC
    -> BEGIN
    ->     DECLARE avgRate FLOAT;
    ->
    ->     SELECT AVG(Rating) INTO avgRate
    ->     FROM appraisal;
    ->
    ->     RETURN avgRate;
    -> END$$
Query OK, 0 rows affected (0.01 sec)

mysql>
mysql> DELIMITER ;
```

Testing:

```
mysql> SELECT AvgRating();
+-------------+
| AvgRating() |
+-------------+
|           5 |
+-------------+
1 row in set (0.00 sec)
```

2. Employee Performance Category

```
mysql> DELIMITER $$
mysql>
mysql> CREATE FUNCTION PerformanceCategory(rate INT)
    -> RETURNS VARCHAR(20)
    -> DETERMINISTIC
    -> BEGIN
    ->     DECLARE result VARCHAR(20);
    ->
    ->     IF rate >= 4 THEN
    ->         SET result = 'Excellent';
    ->     ELSEIF rate >= 3 THEN
    ->         SET result = 'Good';
    ->     ELSE
    ->         SET result = 'Needs Improvement';
    ->     END IF;
    ->
    ->     RETURN result;
    -> END$$
Query OK, 0 rows affected (0.01 sec)
```

```
mysql> SELECT Emp_ID, Rating, PerformanceCategory(Rating) from appraisal;
+--------+--------+----------------------------+
| Emp_ID | Rating | PerformanceCategory(Rating) |
+--------+--------+----------------------------+
|    101 |      5 | Excellent                  |
+--------+--------+----------------------------+
1 row in set (0.00 sec)
```

3. Total Appraisal of an Employee

```
mysql> DELIMITER $$
mysql>
mysql> CREATE FUNCTION TotalAppraisals(empId INT)
    -> RETURNS INT
    -> DETERMINISTIC
    -> BEGIN
    ->     DECLARE total INT;
    ->
    ->     SELECT COUNT(*) INTO total
    ->     FROM appraisal
    ->     WHERE Emp_ID = empId;
    ->
    ->     RETURN total;
    -> END$$
Query OK, 0 rows affected (0.01 sec)

mysql>
mysql> DELIMITER ;
```

Testing:

```
mysql> SELECT Emp_ID, TotalAppraisals(Emp_ID) AS Total_Appraisals from employee;
+--------+------------------+
| Emp_ID | Total_Appraisals |
+--------+------------------+
|    101 |                1 |
|    102 |                0 |
|    103 |                0 |
+--------+------------------+
3 rows in set (0.00 sec)
```